The Nyquist Criterion

A Frequency Response Analysis Technique

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Nyquist Plot or Polar Plot

- Nyquist Plots were invented by Nyquist - who worked at Bell Laboratories, the premiere technical organization in the U.S. at the time.
- Nyquist Plots are a way of showing frequency responses of linear systems.
- There are several ways of displaying frequency response data, including Bode' plots and Nyquist plots.
- Bode' plots use frequency as the horizontal axis and use two separate plots to display amplitude and phase of the frequency response.
- Nyquist plots display both amplitude and phase angle on a single plot, using frequency as a parameter in the plot.
- Nyquist plots have properties that allow you to see whether a system is stable or unstable.

Nyquist Plot

- A Nyquist plot is a polar plot of the frequency response function of a linear system.
- That means a Nyquist plot is a plot of the transfer function, G(s) with s = j**ω**. That means you want to plot G(j **ω**).
- G(j **ω**) is a complex number for any angular frequency, **ω**, so the plot is a plot of complex numbers.
- The complex number, G(j **ω**), depends upon frequency, so frequency will be a parameter if you plot the imaginary part of G(j **ω**) against the real part of G(j **ω**).

Sketch the Polar plot of Frequency Response

To sketch the polar plot of $G(j\omega)$ for the entire range of frequency ω, i.e., from 0 to infinity, there are four key points that usually need to be known:

1) The start of plot where $\omega = 0$,
2) The end of plot where $\omega = \infty$,
3) Where the plot crosses the real axis, i.e., $\mathrm{Im}(G(j\omega)) = 0$, and
4) Where the plot crosses the imaginary axis, i.e., $\mathrm{Re}(G(j\omega)) = 0$.

Problem-1: Polar Plot of Integrator

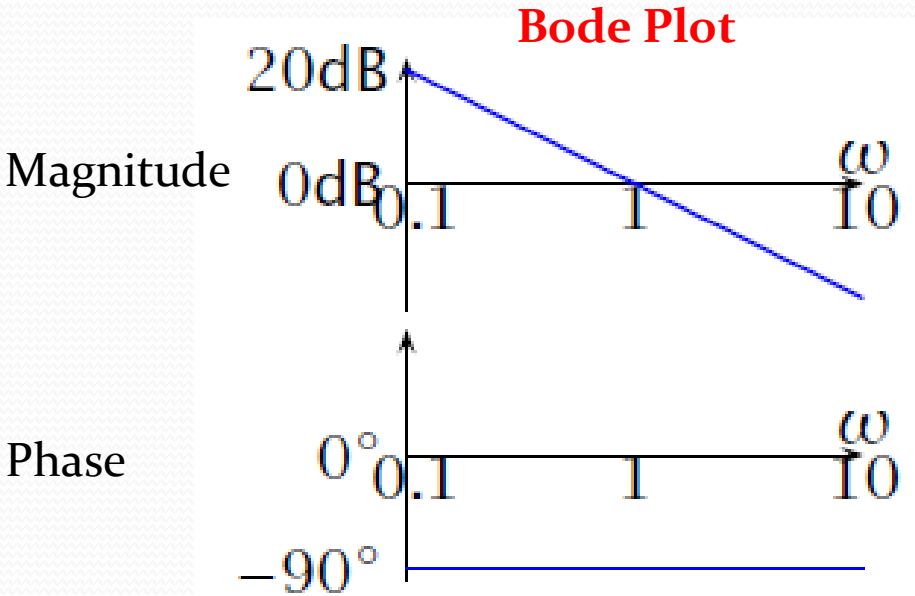
Consider a first order system, $G(s) = \dfrac{1}{s}$

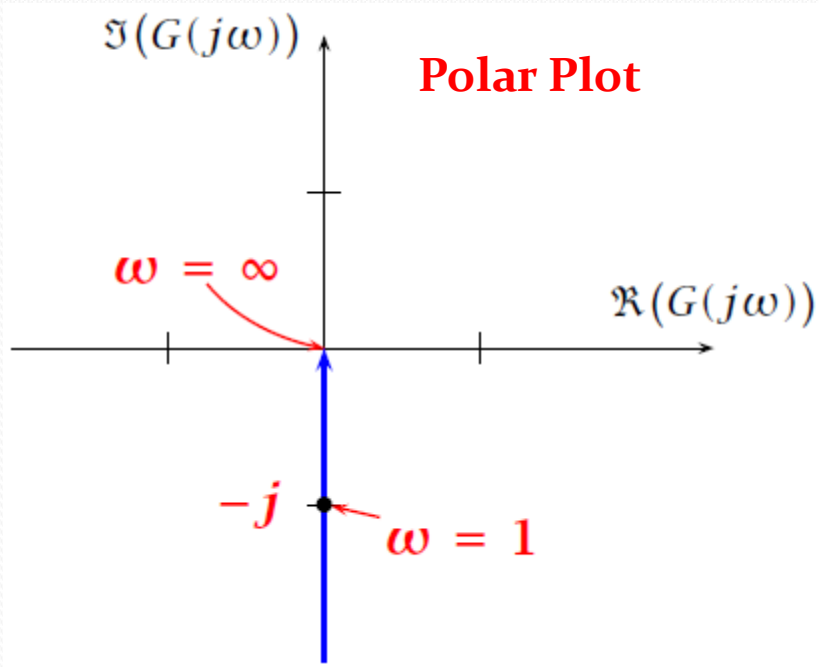
Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing $s = j\omega$:

$$G(j\omega) = \frac{1}{j\omega} = \frac{-j}{\omega}$$

The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as; $|G(j\omega)| = 1/\omega$

The **phase** of $G(j\omega)$, denoted by, ϕ, is obtained as; $\angle G(j\omega) = -90^\circ$





Problem-2: Polar Plot of First Order System

Consider a first order system $G(s) = \dfrac{1}{1+sT}$ where T is the time constant.

Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing $\mathbf{s = j\omega}$:

$$G(j\omega) = \frac{1}{1+j\omega T}$$

The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as;

$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2 T^2}}$$

The **phase** of $G(j\omega)$, denoted by, ϕ , is obtained as;

$$\angle G(j\omega) = -\arctan(\omega T)$$

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The start of plot where ω = 0

$$|G(j\omega)| = \frac{1}{\sqrt{1+0}} = 1, \qquad \phi = \tan^{-1}\left(\frac{0}{1}\right) = 0$$

The end of plot where ω = ∞

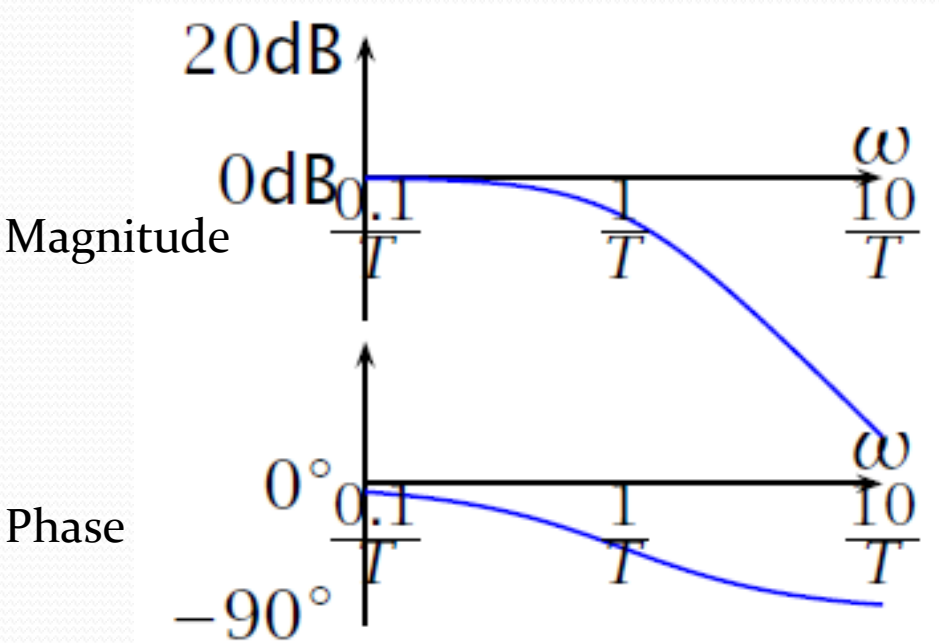
$$|G(j\omega)| = \frac{1}{\sqrt{1+\infty}} = 0, \qquad \phi = \tan^{-1}\left(\frac{-\infty}{1}\right) = -90^\circ$$

The mid part of plot where ω = 1/T

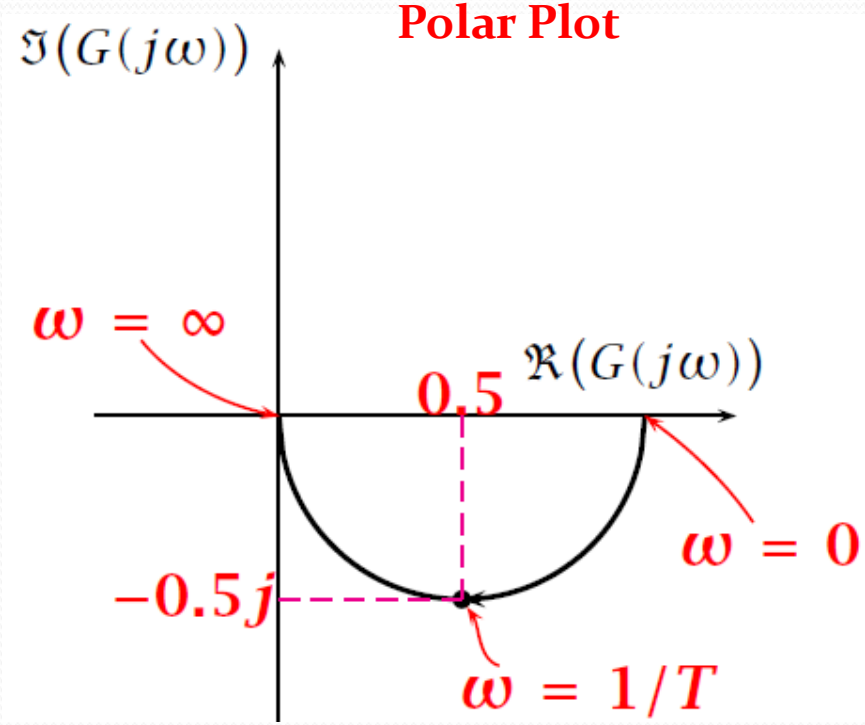
$$|G(j\omega)| = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}, \qquad \phi = \tan^{-1}\left(\frac{-1}{1}\right) = -45^\circ$$

	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
$\omega = 0$	1	0
$\omega = \frac{1}{T}$	$\frac{1}{\sqrt{2}}$	$-45°$
$\omega = \rightarrow \infty$	0	$-90°$

Bode Plot



Polar Plot



Problem-3: Polar Plot of Second Order System

Consider a second order system $G(s) = \dfrac{1}{(1+sT_1)(1+sT_2)}$ where T is the time constant.

Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing **s = jω**:

$$G(j\omega) = \frac{1}{(1+j\omega T_1)(1+j\omega T_2)}$$

The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as;

$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2 T_1^2}\sqrt{1+\omega^2 T_2^2}}$$

The **phase** of $G(j\omega)$, denoted by, ϕ , is obtained as;

$$\angle G(j\omega) = -\arctan(\omega T_1) - \arctan(\omega T_2)$$

The start of plot where ω = 0

$$|G(j\omega)| = \frac{1}{\sqrt{1+0}\sqrt{1+0}} = 1$$

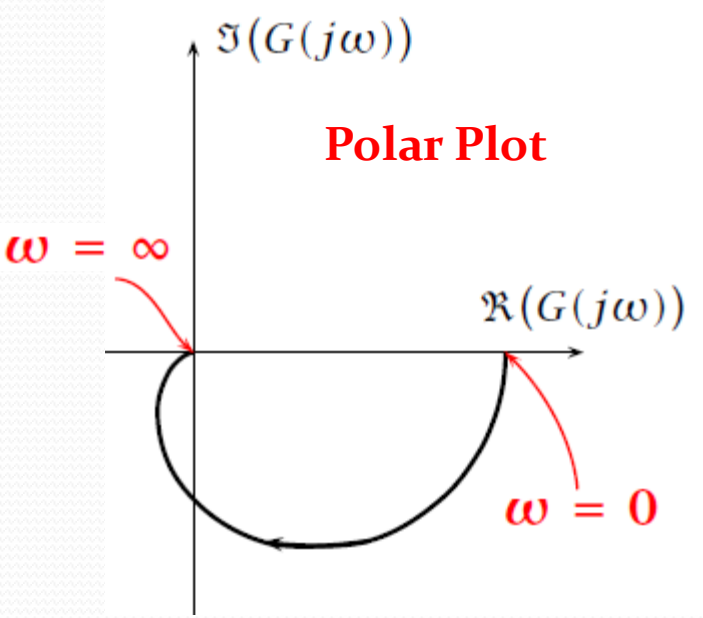
$$\angle G(j\omega) = -\tan^{-1}(0) - \tan^{-1}(0) = 0°$$

The end of plot where ω = ∞

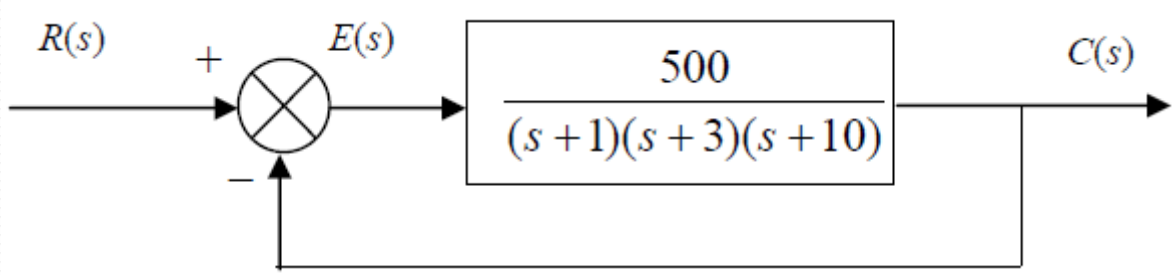
$$|G(j\omega)| = \frac{1}{\sqrt{\infty}\sqrt{\infty}} = 0$$

$$\angle G(j\omega) = -\tan^{-1}(\infty) - \tan^{-1}(\infty) = -90° - 90° = -180°$$

	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
$\omega = 0$	1	0
$\omega = \rightarrow \infty$	0	$-180°$



Problem-4: Sketch the Nyquist diagram for the system shown in the following figure, and then determine the system stability using the Nyquist criterion.



The open-loop transfer function: $G(s)H(s) = \dfrac{500}{(s+1)(s+3)(s+10)}$.

Replacing s with $j\omega$ yields the frequency response of $G(s)H(s)$, i.e.,

$$G(j\omega)H(j\omega) = \frac{500}{(j\omega+1)(j\omega+3)(j\omega+10)} = \frac{500}{(-14\omega^2+30)+j(43\omega-\omega^3)}$$

$$= 500\frac{(-14\omega^2+30)-j(43\omega-\omega^3)}{(-14\omega^2+30)^2+(43\omega-\omega^3)^2} \longrightarrow \text{(a)}$$

Magnitude response:

$$|G(j\omega)H(j\omega)| = \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2} = \frac{500}{\sqrt{(-14\omega^2+30)^2 + (43\omega - \omega^3)^2}}$$

Phase response:

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\left(\frac{\mathrm{Im}}{\mathrm{Re}}\right) = \tan^{-1}\left(\frac{-(43\omega - \omega^3)}{-14\omega^2 + 30}\right)$$

Now that we have expressions for the magnitude and phase of the frequency response, we can sketch the polar plot using the **4 key points**.

Point 1: The start of plot where ω = 0

$$|G(0)H(0)| = \frac{500}{\sqrt{(30)^2}} = 16.67$$

$$\angle G(0)H(0) = \tan^{-1}\frac{0}{30} = 0^\circ$$

Point 2: The end of plot where ω = ∞

$$|G(\infty)H(\infty)| = \frac{500}{\sqrt{\infty}} = 0 \qquad \angle G(\infty)H(\infty) = \tan^{-1}\frac{\infty^3}{30} = -3 \times 90° = -270°$$

Point 3: Where the plot crosses the real axis, i.e., **Im($G(j\omega)$) = 0**

Take the imaginary part of equation (a), and put equal to zero, to get the value of frequency ω at the interception of real axis.

$$\frac{-(43\omega-\omega^3)}{(-14\omega^2+30)^2+(43\omega-\omega^3)^2} = 0 \Rightarrow \begin{cases} 43\omega-\omega^3 = 0 \\ \omega = \infty \end{cases} \Rightarrow \quad \omega = 0 \quad \text{and} \quad \omega = 6.56\,\text{rad/s}$$

Point 4: Where the plot crosses the imaginary axis, Re($G(j\omega)$) = 0

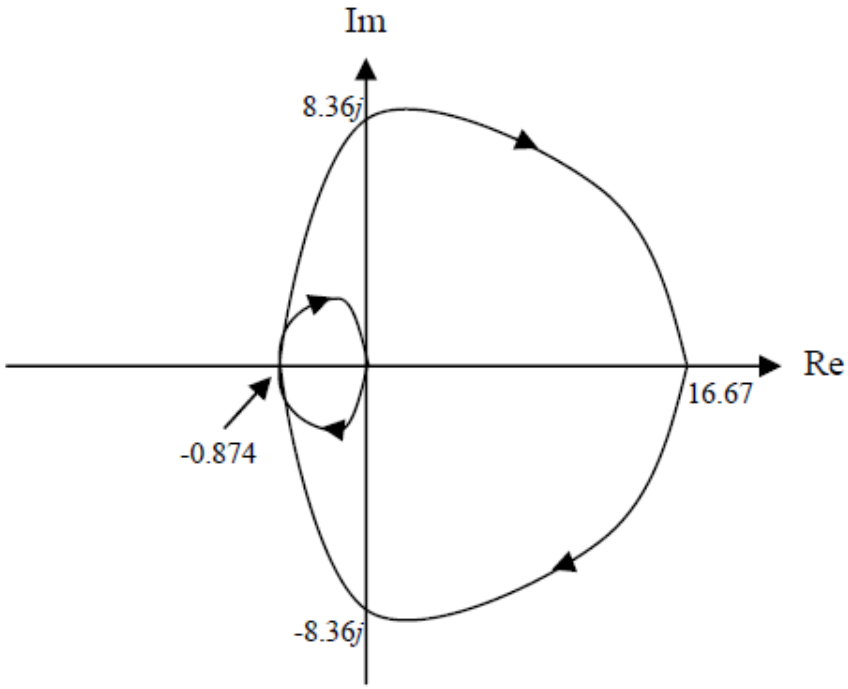
Take the real part of equation (a), and put equal to zero, to get the value of frequency ω at the interception of imaginary axis.

$$\frac{-14\omega^2+30}{(-14\omega^2+30)^2+(43\omega-\omega^3)^2}=0 \Rightarrow \begin{cases} -14\omega^2+30=0 & \Rightarrow \quad \omega=1.46\,\text{rad/s} \\ \omega=\infty \end{cases}$$

Key Points of the polar plot:

	$\lvert GH \rvert$	$\angle GH$
$\omega=0$	16.67	0
$\omega=\infty$	0	-270°
Cross Re: $\omega=0$ $\omega=\infty$ $\omega=6.56$ rad/s	 See above 0.874	 See above -180°
Cross Im: $\omega=0$ $\omega=\infty$ $\omega=1.46$ rad/s	 See above 8.36	 See above -90°

Nyquist diagram



Problem-5: Sketch the polar plot for the following transfer function.

$$G(s) = \frac{10}{s(s+1)(s+5)}$$

Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing **s = jω**, and then Multiply both numerator and denominator by the conjugate of denominator.

$$G(j\omega) = \frac{10}{j\omega(j\omega+1)(j\omega+5)}$$

$$= \frac{10}{(-\omega^2 + j\omega)(j\omega+5)}$$

$$= \frac{10}{(-j\omega^3 - 5\omega^2 - \omega^2 + 5j\omega)}$$

$$= \frac{10}{-6\omega^2 + j(5\omega - \omega^3)} * \frac{-6\omega^2 - j(5\omega - \omega^3)}{-6\omega^2 - j(5\omega - \omega^3)}$$

$$G(j\omega) = \frac{-60\omega^2 - j10\omega(5-\omega^2)}{36\omega^4 - (5\omega - \omega^3)^2} \longrightarrow \text{(a)}$$

Point 1: The start of plot where ω = 0

At frequency ω = 0, we only observe the most significant terms that take the effect.

Magnitude at ω = 0:

$$G(j\omega)\Big|_{\omega=0} = \frac{10}{5j\omega}\Bigg|_{\omega=0} = \frac{2}{j\omega}\Bigg|_{\omega=0} = \infty$$

Phase at ω = 0:

$$\angle G(j\omega)\Big|_{\omega=0} = \lim_{\omega\to 0} \angle \frac{2}{j\omega} = -90^\circ$$

Point 2: The end of plot where ω = ∞

At frequency ω = ∞, we shall look at the most significant term that takes effect when the frequency approaches infinity.

Magnitude at ω = ∞:

$$\left|G(j\omega)\right|_{\omega\to\infty} = \lim_{\omega\to\infty}\left|\frac{10}{(j\omega^3)}\right| = \lim_{\omega\to\infty}\frac{10}{\omega^3} = 0$$

Phase at ω = ∞:

$$\angle G(j\omega)\,|_{\omega\to\infty} = \angle \lim_{\omega\to\infty}\left[\frac{10}{(j\omega)^3}\right] = -270^\circ$$

Point 3: Where the plot crosses the real axis, i.e., **Im($G(j\omega)$) = 0**

Take the imaginary part of equation (a), and put equal to zero, to get the value of frequency ω at the interception of real axis.

$$\Rightarrow -\frac{10\omega(5-\omega^2)}{36\omega^4+(5\omega-\omega^3)^2}=0$$

$$\Rightarrow 10(5-\omega^2)=0$$

$$\Rightarrow \omega^2=5$$

$$\Rightarrow \omega=\sqrt{5}$$

Therefore, the intersection point between the polar plot and the real axis, when $\boldsymbol{\omega} = \sqrt{\mathbf{5}}$, is located at;

$$G(j\omega)\big|_{\omega=\sqrt{5}}=-\frac{1}{3}$$

Point 4: Where the plot crosses the imaginary axis, Re($G(j\omega)$) = 0

Take the real part of equation (a), and put equal to zero, to get the value of frequency ω at the interception of imaginary axis.

$$\Rightarrow \quad -\frac{60\omega^2}{36\omega^4 + (5\omega - \omega^3)^2} = 0$$

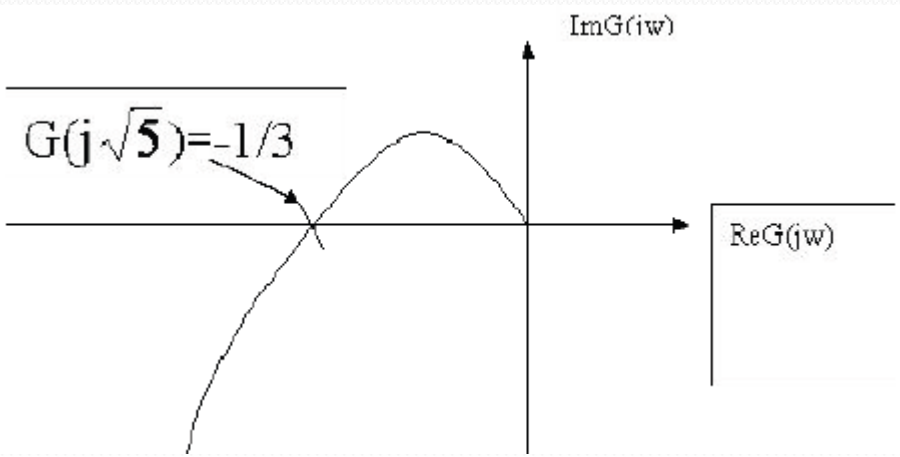
$$\Rightarrow \quad \omega = \infty$$

Therefore, the intersection point between the polar plot and the imaginary axis is when $\omega = \infty$ is located at;

$$G(j\infty) = 0$$

	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
$\omega = 0$	∞	-90°
$\omega = \infty$	0	-270°
Cross Re: $\omega = \sqrt{5}$	$-\frac{1}{3}$	
Cross Img: $\omega = \infty$	0	

Polar Plot



Problem-6: Sketch the polar plot for the following transfer function.

$$GH(s) = \frac{1}{s^4(s+p)} \qquad p > 0$$

Representing $G(s)H(s)$ in the frequency response form $G(j\omega)H(j\omega)$ by replacing $\mathbf{s = j\omega}$:

$$GH(j\omega) = \frac{1}{j^4\omega^4(j\omega + p)}$$

The **magnitude** of $GH(j\omega)$ i.e., $|GH(j\omega)|$, is obtained as;

$$|GH(j\omega)| = \frac{1}{\omega^4\sqrt{\omega^2 + p^2}}$$

The **phase** of $GH(j\omega)$ denoted by, φ, is obtained as;

$$\varphi = \angle GH(0) = -\tan^{-1}\left(\frac{\omega}{p}\right)$$

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Point 1: **The start of plot where ω = 0**

$$|GH(0)| = \frac{1}{0\sqrt{0+p^2}} = \infty$$

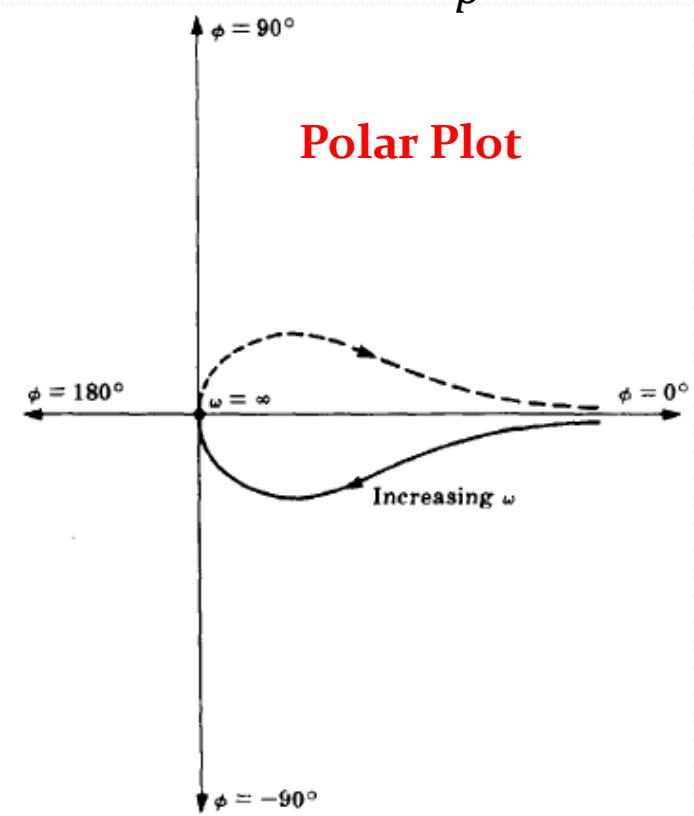
$$\varphi = \angle GH(0) = -\tan^{-1}\frac{0}{p} = 0^{\circ}$$

Point 2: **The end of plot where ω = ∞**

$$|GH(\infty)| = \frac{1}{\infty\sqrt{\infty+p^2}} = 0$$

$$\varphi = \angle GH(\infty) = -\tan^{-1}\frac{\infty}{p} = -90^{\circ}$$

	|GH(jω)|	∠GH(jω)
ω = 0	∞	0°
ω = ∞	0	-90°



Evaluation of a Complex Function via Vectors

PROBLEM: Given

$$F(s) = \frac{(s+1)}{s(s+2)} \tag{8.7}$$

find $F(s)$ at the point $s = -3 + j4$.

SOLUTION: The problem is graphically depicted in Figure 8.3, where each vector, $(s+\alpha)$, of the function is shown terminating on the selected point $s = -3 + j4$. The vector originating at the zero at -1 is

$$\sqrt{20}\angle 116.6^\circ \tag{8.8}$$

The vector originating at the pole at the origin is

$$5\angle 126.9^\circ \tag{8.9}$$

The vector originating at the pole at -2 is

$$\sqrt{17}\angle 104.0^\circ \tag{8.10}$$

Substituting Eqs. (8.8) through (8.10) into Eqs. (8.5) and (8.6) yields

$$M\angle\theta = \frac{\sqrt{20}}{5\sqrt{17}}\angle 116.6^\circ - 126.9^\circ - 104.0^\circ = 0.217\angle -114.3^\circ \tag{8.11}$$

as the result for evaluating $F(s)$ at the point $-3 + j4$.

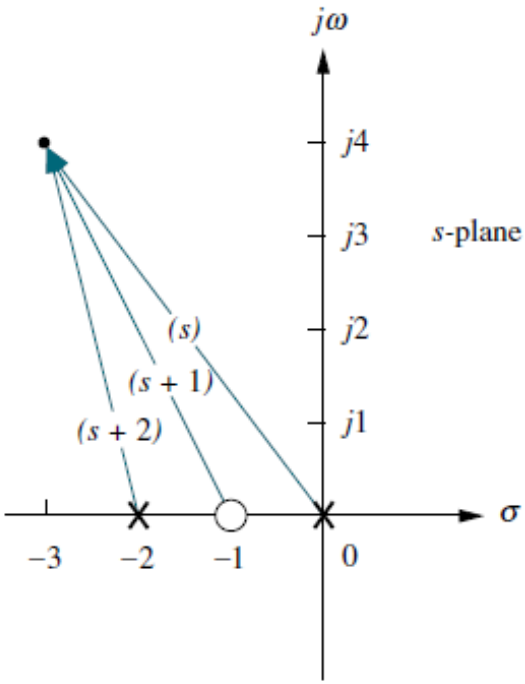


FIGURE 8.3 Vector representation of Eq. (8.7)

PROBLEM: Given

$$F(s) = \frac{(s+2)(s+4)}{s(s+3)(s+6)}$$

find $F(s)$ at the point $s = -7 + j9$ the following ways:

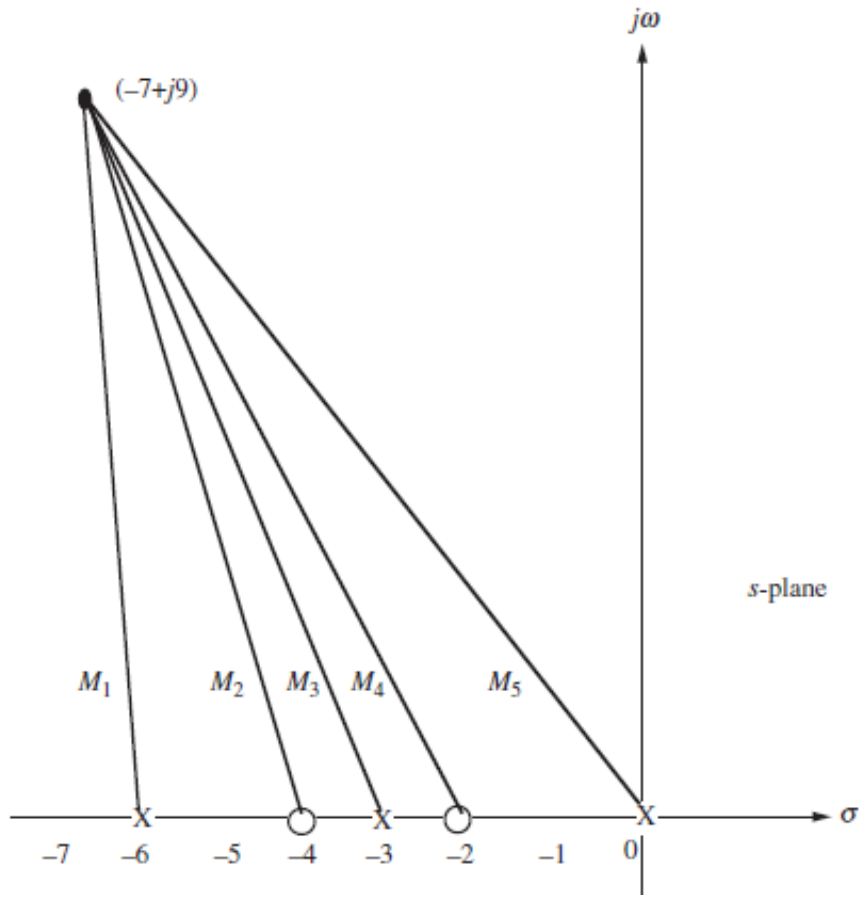
a. Directly substituting the point into $F(s)$

b. Calculating the result using vectors

a. $$F(-7+j9) = \frac{(-7+j9+2)(-7+j9+4)0.0339}{(-7+j9)(-7+j9+3)(-7+j9+6)} = \frac{(-5+j9)(-3+j9)}{(-7+j9)(-4+j9)(-1+j9)}$$

$$= \frac{(-66-j72)}{(944-j378)} = -0.0339 - j0.0899 = 0.096 < -110.7^\circ$$

b. The arrangement of vectors is shown as follows:



From the diagram,

$$F(-7+j9) = \frac{M_2 M_4}{M_1 M_3 M_5} = \frac{(-3+j9)(-5+j9)}{(-1+j9)(-4+j9)(-7+j9)} = \frac{(-66-j72)}{(944-j378)}$$

$$= -0.0339 - j0.0899 = 0.096 <; -110.7^\circ$$

Four Important Points for Derivation the Nyquist Criterion

1. The relationship between the poles of $1 + G(s)H(s)$ and the poles of $G(s)H(s)$;

1. The relationship between the zeros of $1 + G(s)H(s)$ and the poles of the closed-loop transfer function, $T(s)$;

2. The concept of mapping points;

3. The concept of mapping contours.

Letting

$$G(s) = \frac{N_G}{D_G}$$

$$H(s) = \frac{N_H}{D_H}$$

we find

$$G(s)H(s) = \frac{N_G N_H}{D_G D_H}$$

$$1 + G(s)H(s) = 1 + \frac{N_G N_H}{D_G D_H} = \frac{D_G D_H + N_G N_H}{D_G D_H}$$

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{N_G D_H}{D_G D_H + N_G N_H}$$

It is conclude that;

1) The poles of 1 + G(s)H(s) are the same as the poles of G(s)H(s), the open-loop system.

2) The zeros of 1 + G(s)H(s) are the same as the poles of T(s), the closed-loop system.

The Concept of Mapping

- If we take a complex number s = x+*j*y on the s-plane and substitute it into a function, F(s), another complex number results. This process is called mapping.

OR

- The term mapping is defined as the substitution of a complex number into a function, F(s), to get another complex number.

- **For example**, substituting **s = 4 + j3** into the function **F(s) = (s^2 +2s + 1)** yields **16+j30**. We say that 4 + j3 maps into 16+j30 through the function s^2 +2s + 1.

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The Concept of Mapping Contours

The collection of points, called a **contour**.

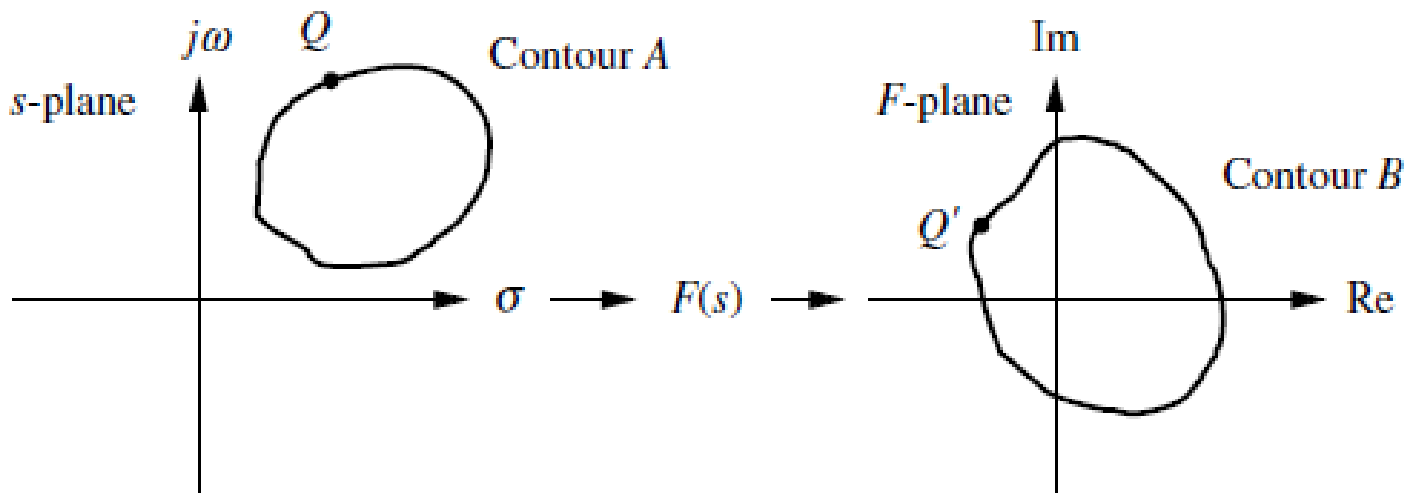
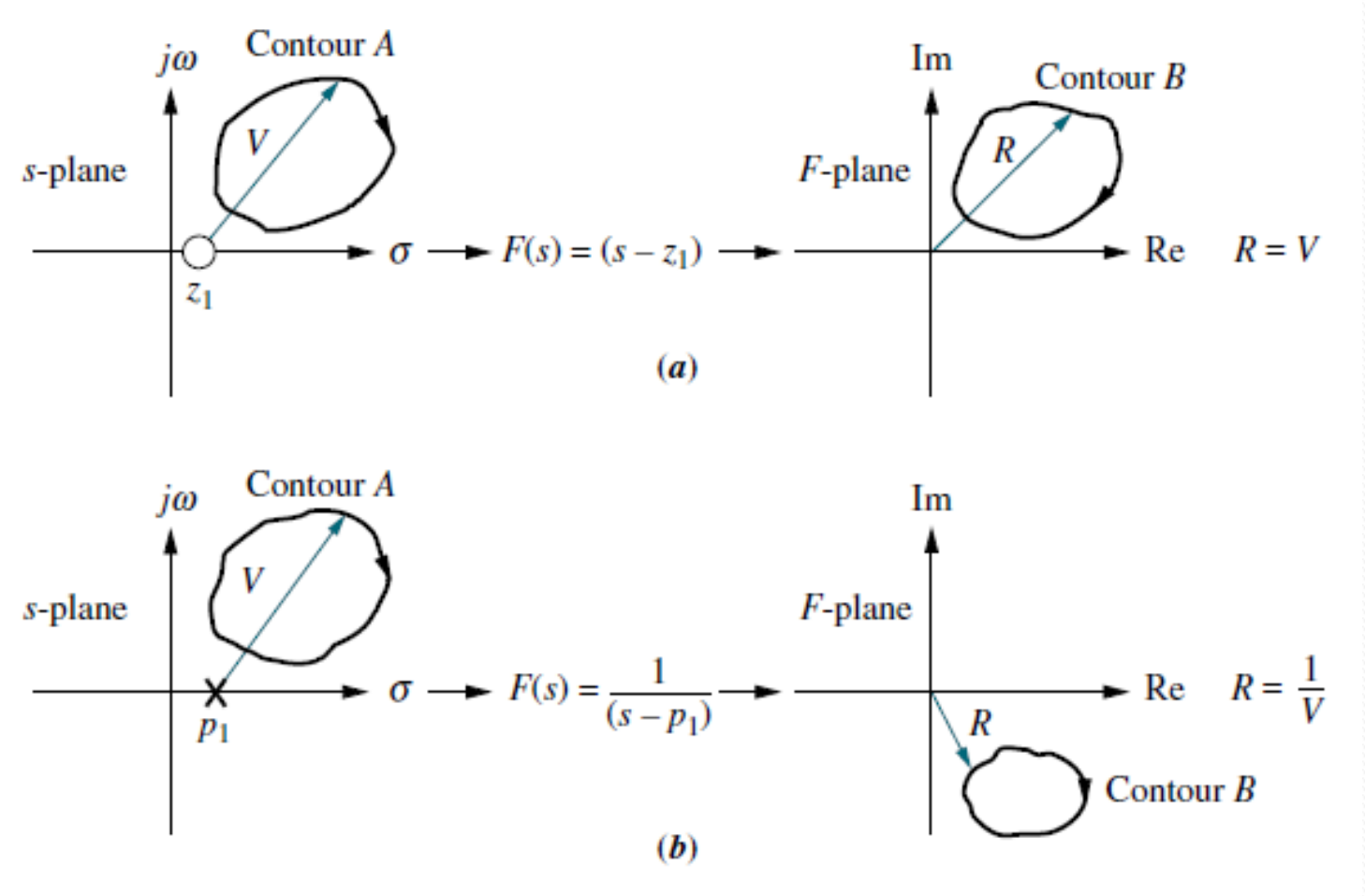


FIGURE 10.21 Mapping contour A through function $F(s)$ to contour B

Contour A can be mapped through $F(s)$ into contour B by substituting each point of contour A into the function $F(s)$ and plotting the resulting complex numbers. For example, point Q in Figure 10.21 maps into point Q' through the function $F(s)$.

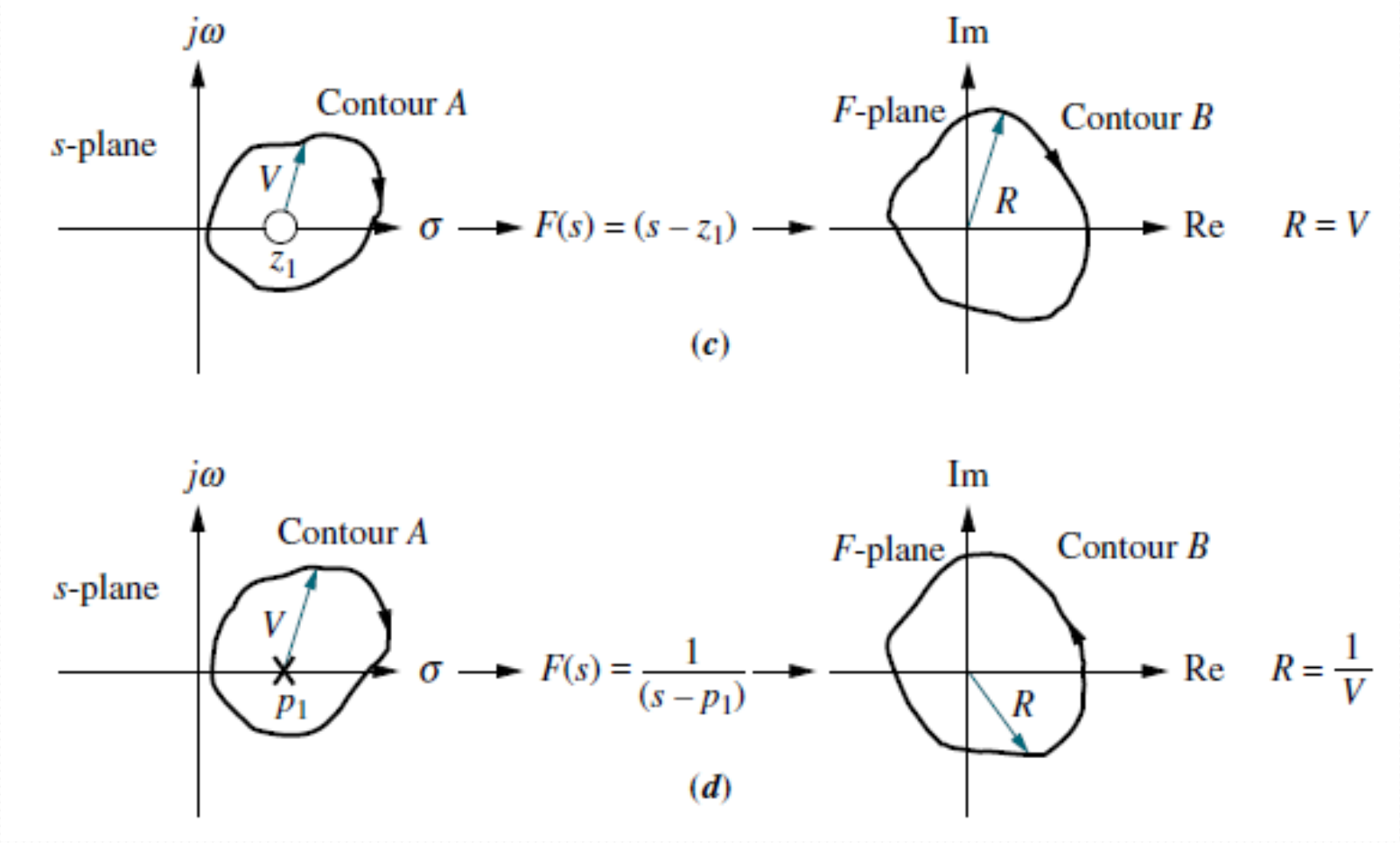
Examples of Contour Mapping

If we assume a clockwise direction for mapping the points on contour **A**, the contour **B** maps a clockwise direction if **F(s)** has just zeros or has just poles that are not encircled by the contour.



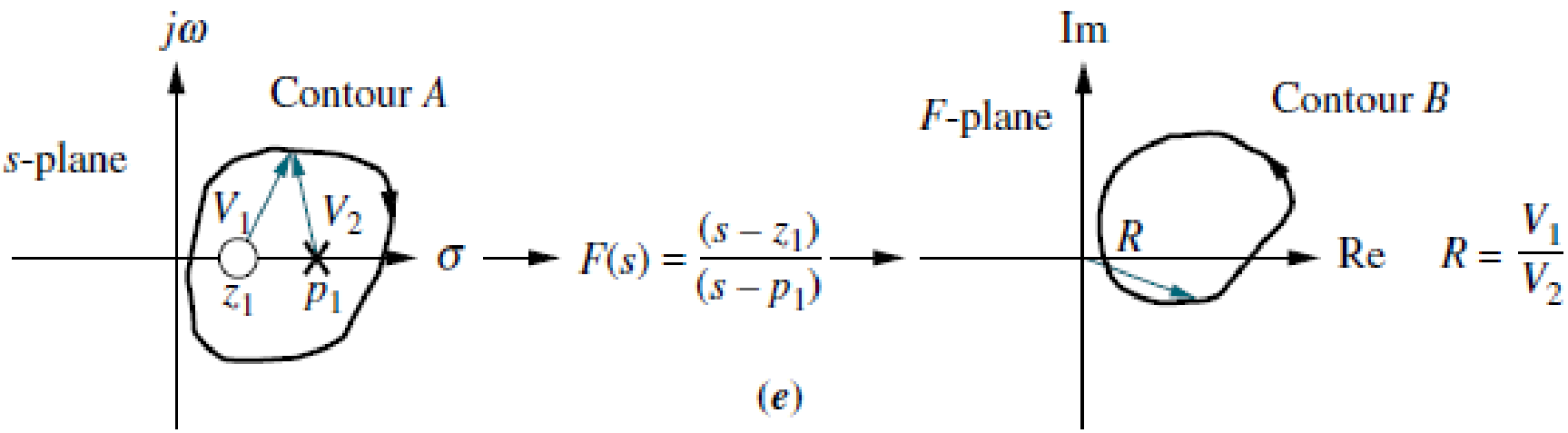
Examples of Contour Mapping

The **contour B** maps in a counter clockwise direction if **F(s)** has just poles that are encircled by the contour, Also, you should verify that, if the pole or zero of **F(s)** is enclosed by **contour A**, the mapping encircles the origin .



Examples of Contour Mapping

In this last case, the pole and zero rotation cancel, and the mapping does not encircle the origin.



THE NYQUIST STABILITY CRITERION

- **A** linear closed-loop ***continuous control system*** is absolutely stable if the roots of the characteristic equation have negative real parts.

- Equivalently, the poles of the closed-loop transfer function, or the zeros of the denominator, ***1 + GH(s),*** of the closed-loop transfer function, must lie in the left-half plane.

- The Nyquist Stability Criterion establishes the number of zeros of ***1 + GH(s)*** in the right-half plane directly from the Nyquist Stability Plot of ***GH(s)***.

Nyquist Stability Criterion

The closed-loop control system whose open-loop transfer function is ***GH*** is stable if and only if

$$N = -P_0 \leq 0$$

Where,

N = total number of **CW** encirclements of the **(-1,0)** point (i.e., ***GH*** = - ***1***) in the *GH*-plane.
P_o = number of poles of ***GH*** in the right-hand-plane RHP for continuous systems.

If ***N*** > 0, the number of zeros Z_o of **1 + GH** in the RHP for continuous systems, is determined by

$$Z_0 = N + P_0$$

If $N \leq 0$ and $P_o = 0$, then the system is absolutely stable if and only if $N = 0$; that is, if and only if the **(-1,0)** point does not lie in the shaded region.

Problem-7: Sketch the polar plot for the following transfer function and check the stability of the system.

$$GH(s) = \frac{1}{s(s+1)}$$

There is one pole at the origin.

Representing $G(s)H(s)$ in the frequency response form $G(j\omega)H(j\omega)$ by replacing **s = jω**:

$$GH(j\omega) = \frac{1}{j\omega(j\omega+1)}$$

The **magnitude** of $GH(j\omega)$ i.e., $|GH(j\omega)|$, is obtained as;

$$|GH(j\omega)| = \frac{1}{\omega\sqrt{\omega^2+1}}$$

The **phase** of $GH(j\omega)$ denoted by, φ, is obtained as;

$$\varphi = \angle GH(0) = -90° - \tan^{-1}\left(\frac{\omega}{1}\right)$$

Point 1: **The start of plot where ω = 0**

$$|GH(0)| = \frac{1}{0\sqrt{0+1}} = \infty$$

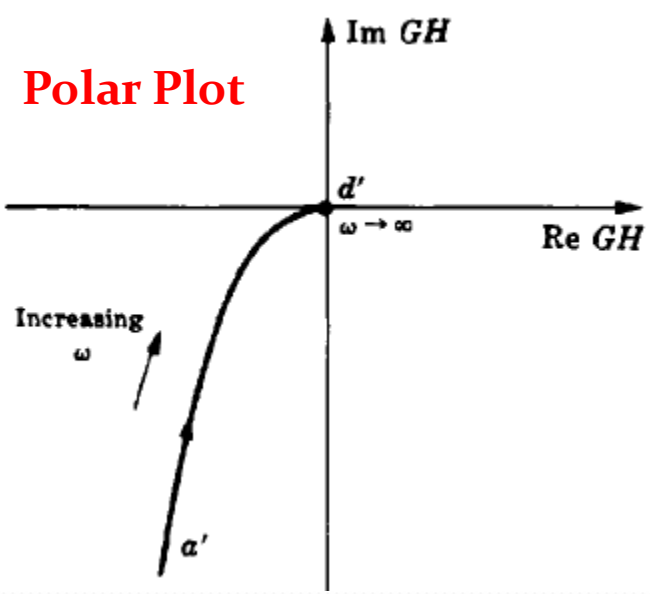
$$\varphi = \angle GH(0) = -90° - \tan^{-1}\frac{0}{1} = -90^{\circ}$$

Point 2: **The end of plot where ω = ∞**

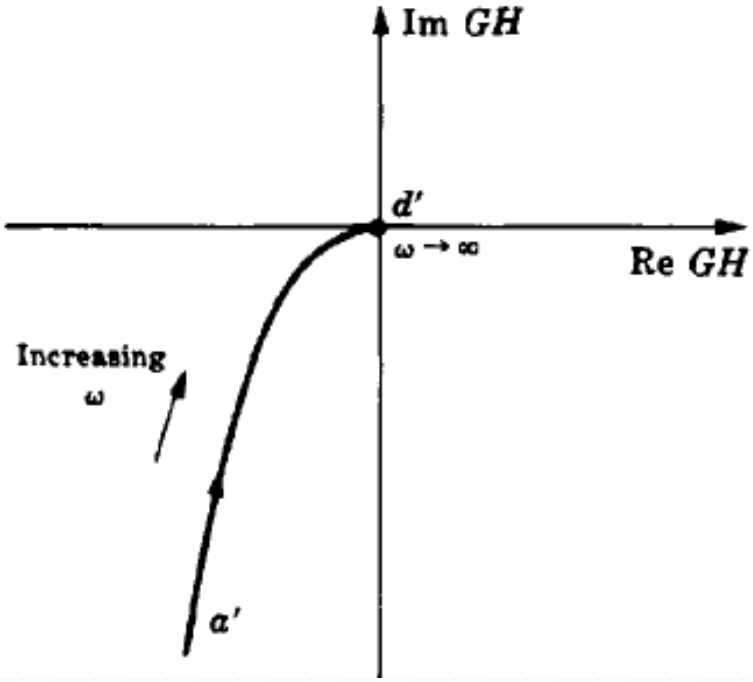
$$|GH(\infty)| = \frac{1}{\infty\sqrt{\infty+1}} = 0$$

$$\varphi = \angle GH(\infty) = -90 - \tan^{-1}\frac{\infty}{p} = -90° - 90° = -180°$$

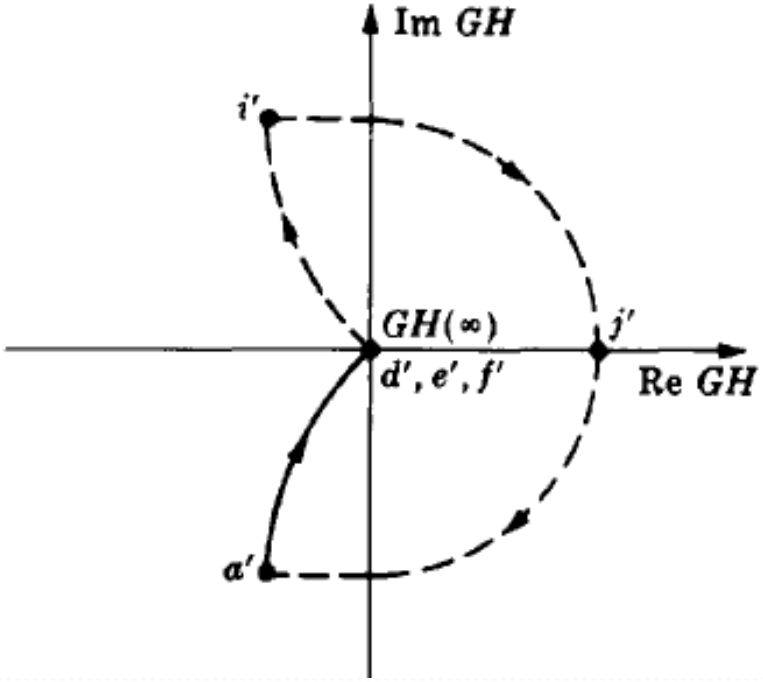
	|GH(jω)|	∠GH(jω)
ω = 0	∞	-90°
ω = ∞	0	-180°



Nyquist or Polar Plot



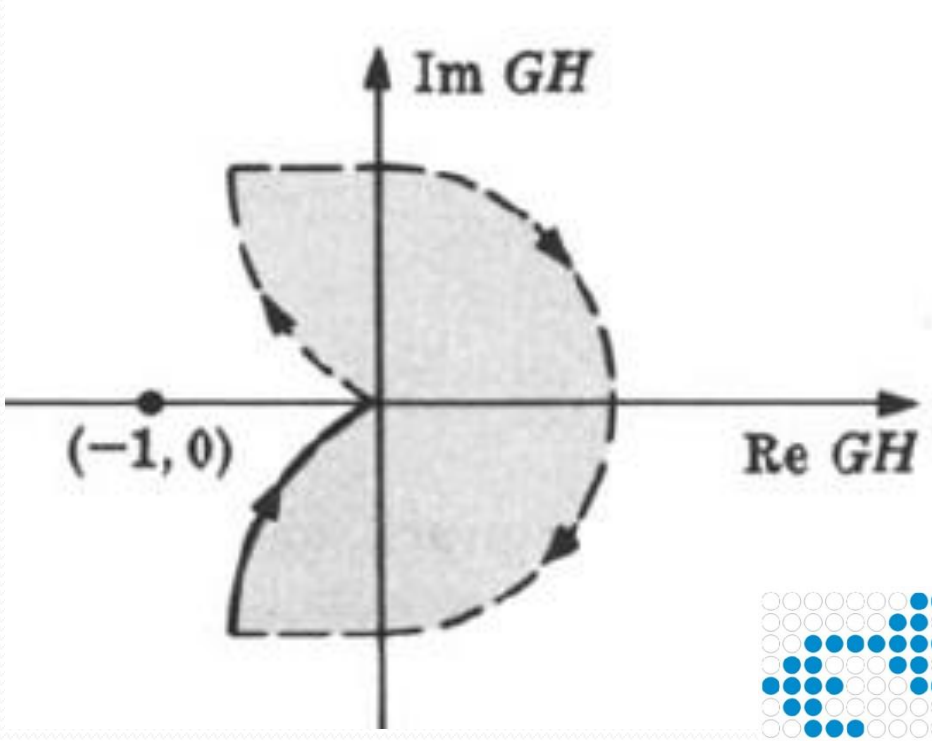
Nyquist Diagram

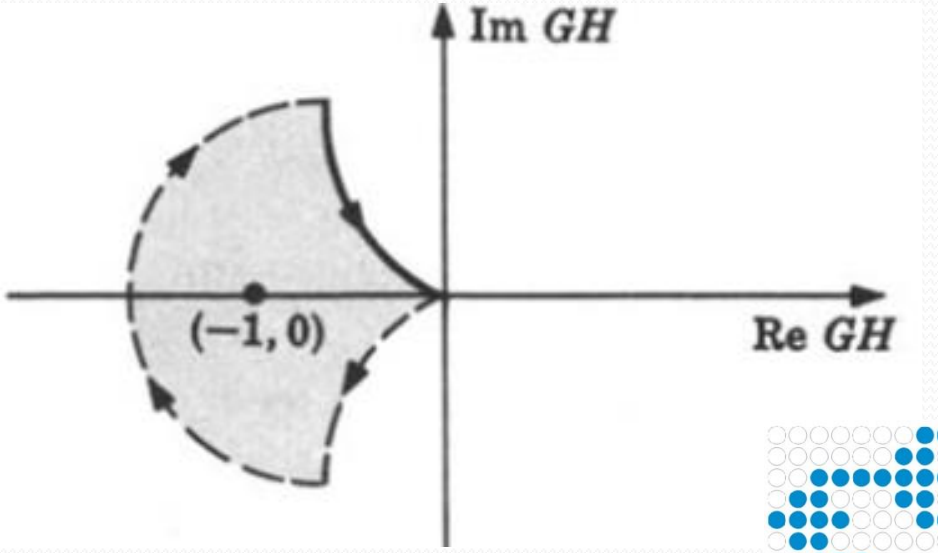


- The region to the right of the contour has been shaded.

- Clearly, the **(-1,0)** point is not in the shaded region; therefore it is not enclosed by the contour and so $N \leq 0$.

- The poles of $GH(s)$ are at $s = 0$ and $s = -1$, neither of which are in the right-hand-plane RHP; hence $P_o = 0$. Thus $N = -P_o = 0$, and the system is absolutely stable.



Problem-8: The Nyquist Stability Plot for $GH(s) = 1/s(s-1)$ is given in the figure below.

- The region to the right of the contour has been shaded and the **(-1,0)** point is enclosed; then $N > 0$. (It is clear that $N = 1$).
-
- The poles of GH are at $s = 0$ and $s = +1$, the latter pole being in the RHP. Hence $P_o = 1$.
- $N \neq P_o$ indicates that the **system is unstable**.
- $Z = N + P_o = 1 + 1 = 2$, therefore the poles of the closed-loop transfer function, or the zeros of the characteristic equation $1+GH$ in the right-hand plane RHP.